

A Benchmark Paper for Cardiac Substructure 3D Reconstruction with Dense Correspondence

IEEE Publication Technology, *Staff, IEEE,*

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Index Terms—Article submission, IEEE, IEEEtran, journal, L^AT_EX, paper, template, typesetting.

I. INTRODUCTION

CARDIAC three-dimensional (3D) reconstruction are essential for preoperative assessment and surgical planning, and provide a key foundation for the development of patient-specific digital heart models. A surface shape is mathematically defined as a continuous 2D manifold embedded in 3D Euclidean space. Cardiac 3D reconstruction aims to recover such surfaces for cardiac structures from various input data representations, including 2D slice-based images, volumetric medical data, and point clouds from segmentation of medical images. Crucially, we seek to reconstruct continuous surfaces while simultaneously preserving point correspondences, enabling shape registration across different scans or time frames for accurate alignment, comparison, and analysis. In this work, we are particularly interested in reconstructing continuous surfaces of cardiac substructures from discrete point cloud representations, as point clouds preserve rich spatial geometric properties and are well aligned with the inherently 3D and dynamic nature of the beating heart.

The problem is technically ill-posed — infinitely many solutions of the underlying, continuous surface may exist given an observed set of discrete points. The challenges become even severer considering that, unlike general surface reconstruction, cardiac reconstruction must account for complex anatomical structures, dynamic motion due to the beating heart, and inter-subject variability, all while preserving point correspondences for accurate registration across different scans or time frames. These difficulties are further exacerbated by the limitations of the input data: the raw volumetric images often have low inter-slice resolution, and the point clouds derived from volumetric segmentation are typically sparse and incomplete, making the reconstruction task even more challenging. All these issues pose the cardiac reconstruction from point clouds as a long-standing challenge that draws continuous efforts from the research community.

cardiac reconstruction papers []/ general surface reconstruction methods [] that can be adapted to cardiac reconstruction.

Less works for cardiac reconstruction because of absence of cardiac 3D reconstruction related dataset. Reasons?. It is therefore highly desirable to establish a benchmark for cardiac substructure surface reconstruction, evaluating both classical approaches and recent deep learning-based methods. Such a systematic comparison would not only highlight the respective strengths and limitations of these methods, but also provide valuable guidance for practitioners in selecting appropriate techniques and inspire future innovations in research.

The present paper aims to TODO

A. Contributions

Placeholder

B. Paper Organization

Placeholder

II. RELATED WORKS

In this section, we give a summary of existing literature on surface reconstruction from either point clouds or other observations. We also summarize existing datasets and benchmarks that have served for advancing the field.

A. Surface reconstruction from

B. Surface reconstruction from point clouds

C. Datasets and benchmarks

III. PROBLEM STATEMENT

3D Cardiac have consistent topologies with limited structure variations. We work on shape variations under 3D reconstruction. However, in shape registration, considering structure variations is necessary(SDF delivers structure variations in different iso-surface.). correspondence construction is challenging, even humans cannot reliably annotate dense correspondences between two cardiac frames. structure variations / motion variations(4D) goal: high-fidelity 3D shapes, build high-quality dense correspondence

$$\begin{aligned} \mathcal{L}_{SDF} = & \sum_i \left(\sum_{p \in \Omega} |f_{\theta}^s(p, z_i) - s| + \right. \\ & \sum_{p \in \Omega} (1 - \langle \nabla f_{\theta}^s(p, z_i), n \rangle) + \\ & \sum_{p \in \Omega} (||\nabla f_{\theta}^s(p, z_i)||_2 - 1) + \\ & \left. \sum_{p \in \Omega} \rho(f_{\theta}^s(p, z_i)) \right) \end{aligned} \quad (1)$$

$$\mathcal{L}_{smooth} = \sum_i \sum_{p \in \Omega} \|\nabla f_{\theta}^v(p, z_i)\|_2 \quad (2)$$

penalizes the spatial gradient of the deformation field along X , Y and Z directions, encouraging smooth deformation and avoid large shape distortion.

consistent loss of sdf and normal in template space

The 3D cardiac reconstruction task considered in this work is to learn a mapping from sparse cardiac geometric observations to dense heart surface representations. Specifically, let $P_t = \{(x_i, l_i)\}_{i=1}^{N_t}$ denote a sparse point cloud at cardiac phase t , where $x_i \in \mathbb{R}^3$ is the spatial coordinate of a point and l_i is its anatomical label when available. The goal is to predict a dense cardiac mesh $M_t = (V_t, F)$, where $V_t \in \mathbb{R}^{K \times 3}$ denotes the vertex coordinates and F denotes the mesh connectivity. For dynamic reconstruction, the task can be further extended from a single frame to a temporal sequence, i.e., $\{P_t\}_{t=1}^T \mapsto \{M_t\}_{t=1}^T$. A practical difficulty is that existing cardiac MRI datasets, including MnMs2-4D, UK Digital Heart, ACDC, and 4DM, do not directly provide unified dense mesh supervision for this sparse-to-dense reconstruction setting. Their original formats are heterogeneous, and typically contain volumetric images, segmentations, contours, or sparse geometric observations rather than standardized dense surface meshes. Therefore, before model learning, we reconstruct and normalize these datasets into a common representation consisting of cardiac MR volumes, sparse point clouds, dense meshes, and corresponding spatial normalization parameters. Based on this unified dataset, the learning objective of our model is to recover a dense and anatomically consistent 3D cardiac shape from sparse input observations. In other words, we aim to learn a function f_{θ} such that $M_t = f_{\theta}(P_t)$, or, in the dynamic setting, $\{M_t\}_{t=1}^T = f_{\theta}(\{P_t\}_{t=1}^T)$. This formulation provides the foundation for both cardiac shape reconstruction and shape generation from sparse cardiac geometric data.

IV. CARDIAC RECONSTRUCTION METHODS

A. Learning-free methods

single-instance geometric reconstruction

B. Learning-based method

1) *3D reconstruction*: AtlasNet shape-coded DeepSDF [1]

2) *3D reconstruction with dense correspondence*: Constructing dense correspondence is necessary during reconstruction. It

Template mesh-based methods explicit, the reconstruction quality depends on the resolution or capacity of the template mesh.

Deformation field-based methods (a template SDF) / Covering dense point correspondence: deformation field-based methods (decompose the DIF into deformation field learning and template field learning) which can learn better dense correspondence? template SDF records all structure variations in a category via its different iso-surface

DIF-Net DIT

deformation field-based methods based on neural differential flow: NDF HNDF SHDF

3) *Other types?*:

V. A CARDIAC RECONSTRUCTION BENCHMARK

As discussed in Section IV, a rich set of methods have been developed for cardiac 3D reconstruction from point clouds extracted via segmentation of medical images. Although these approaches have demonstrated promising performance, their advantages and limitations under different clinical and imaging conditions remain insufficiently understood, mainly due to the lack of a comprehensive benchmark for cardiac reconstruction. Existing studies are often evaluated on limited datasets and heterogeneous experimental settings, making fair comparison difficult. In this work, we present a dedicated benchmark for cardiac 3D reconstruction by constructing diverse datasets and standardized evaluation protocols covering multiple anatomical structures and reconstruction challenges (e.g. high quality reconstruction with less sampled points) and so on. Then benchmark contains several subsets, UKBio(what task, what data),...

After data processing, what does the processed data look like?(e.g. biventricular mesh?) and how many data we can use for 3D reconstruction.

We prepare the SDF samples for each mesh, which consists of 3D points and their SDF values. While SDF can be computed through a distance transform for watertight cardiac shapes we processed. To prepare data, we start by normalizing each mesh to a unit sphere and sampling 500,000 spatial points x 's: we sample more aggressively near the surface of the object as we want to capture a more detailed SDF near the surface.

represent training data / unseen shape reconstruction / shape registration / learn smooth and complete shape embedding space from which we can sample new shapes (can be used in 4D case).

A. UK Digital Heart-Static Biventricular Meshes

1) *Data Collection*: The UK Digital Heart Project [?] provides high-resolution 3D bSSFP cine sequences from 1,331 healthy adults at end-diastole (ED) and end-systole (ES). Ground-truth segmentation masks at $1.2 \times 1.2 \times 2$ mm resolution are provided for the left ventricular blood pool (LV), myocardium (Myo), and right ventricular blood pool (RV).

2) Data Processing Pipeline:

a) *Mask-to-Mesh Conversion*: Each segmentation mask undergoes morphological pre-processing (opening/closing to remove noise and fill holes), followed by Marching Cubes surface extraction. The resulting mesh is then refined via Poisson surface reconstruction, QEM-based edge collapse simplification, isotropic remeshing, and Laplacian smoothing to produce a watertight biventricular surface mesh. The NIFTI affine matrix is saved alongside each mesh to preserve the mapping from voxel to physical coordinates.

b) Simulated CMR Slicing.: To obtain contour point clouds consistent with real clinical acquisitions, we compute cutting planes corresponding to the standard cardiac MRI views: one short-axis (SAX) stack and three long-axis views (2-chamber, 3-chamber, and 4-chamber). The mesh is intersected with each plane, and the resulting contours are resampled to exactly 5,000 points using farthest point sampling (FPS).

c) Canonical Pose Alignment and Normalisation.: An SE(3) rigid transformation aligning each subject's mesh to the SSM reference frame is computed from three anatomical features: the LV centroid, the short-axis normal vector, and the LV-to-septum direction. The aligned vertices are then normalised into a unit sphere and an axis-aligned bounding box (AABB), yielding the canonical representation used for all subsequent SDF sampling.

d) SDF Sampling.: For each watertight mesh in canonical space, signed distance field (SDF) values are computed at densely sampled query points using Kaolin [?], yielding per-point supervision signals $[x, y, z, \text{sdf}_{\text{LV}}, \text{sdf}_{\text{RV}}]$.

B. M&Ms-2-4D Biventricular Mesh Sequences

1) Data Collection.: The M&Ms-2 dataset [?] provides multi-disease, multi-centre short-axis (SAX) and four-chamber long-axis (LAX) cine MRI from 360 subjects, with manual segmentations at ED and ES only.

2) Data Processing Pipeline:

a) Temporal Decomposition and Phase Alignment.:

Subjects for which the LAX and SAX sequences differ in frame count are excluded. The ED frame is identified by pixel-level comparison against the provided reference volume and is re-indexed as frame 0 via the cyclic mapping $\hat{t} = (t - t_{\text{ED}}) \bmod T$. The 4D volume is then decomposed into T independent 3D NIfTI frames.

b) Automatic Segmentation of Intermediate Frames.:

For all frames other than ED and ES, segmentation masks are predicted by separate 2D nnU-Net [?] models trained on the two annotated phases under five-fold cross-validation, independently for the LAX and SAX views.

c) Contour Extraction and Short-Axis Correction.: Per-frame LV and RV contour points are extracted from the complete mask sequences using Marching Squares and B-spline smoothing. Slice-wise breathing-motion misalignment in the SAX stack is corrected using long-axis geometric constraints. All contour coordinates are mapped to physical space (mm) via the NIfTI affine transformation.

d) 4D Statistical Shape Model Fitting.: An SSM prior is used to reconstruct topologically consistent surface meshes across the full cardiac cycle. Starting from a Sim(3) initialisation at the ED frame, the SSM jointly optimises a shared shape code θ (PCA coefficients) and per-frame rigid pose parameters, regularised by a temporal Laplacian smoothness term, following the framework of [?].

e) Mesh Repair.: The LV and RV submeshes are processed independently to remove topological defects; any mesh failing the watertightness criterion after repair is discarded. The canonical pose and SDF sampling then follow the same procedure as in Section ??.

C. 4DM-External LV Prior

The 4DM dataset [?] contains 4D left-ventricular surface meshes from 25 healthy subjects. Since the data are provided as pre-built OBJ meshes with companion pose matrices ($P.txt$), no surface reconstruction is required. We convert all meshes and contour point clouds from OBJ to PLY format, assign per-layer labels by projecting points onto the short-axis normal, and compute normalisation parameters in two coordinate frames: the shared SSM space (for cross-dataset consistency) and the native 4DM space (for direct comparison with the original method). The 4DM mean LV model (`LVM_ED_mean.obj`) additionally serves as the reference shape for normalising ACDC data into the 4DM space.

D. ACDC — Evaluation Point Clouds

The ACDC dataset [?] provides 150 multi-pathology SAX cine sequences with manual ED/ES segmentations. Because its low spatial resolution (1.5–3.5 mm in-plane) and the absence of long-axis views preclude high-quality mesh reconstruction, we use ACDC exclusively for evaluation. Label indices are remapped to match our convention ($\{\text{RV:1, Myo:2, LV:3}\} \rightarrow \{\text{LV:1, RV:3}\}$), and the voxel spacing is used as a surrogate affine in lieu of a true spatial transform. Since the low resolution systematically underestimates ventricular scale, pose alignment is performed via Sim(3) similarity transform (rather than rigid SE(3)), with the scale fixed to the SSM template's prior size. For subjects where LV-only evaluation against the 4DM coordinate system is required, an additional Sim(3) transform into the 4DM space is computed using the mean 4DM LV model as reference.

VI. EXPERIMENTAL SET-UP FOR BENCHMARKING EXISTING SURFACE RECONSTRUCTION METHODS

With the benchmark prepared in Section V, we aim to empirically compare existing surface reconstruction methods, and identify their strengths and weaknesses in cardiac reconstruction under various conditions (e.g. studies on data efficiency or shape code representations). We expect that such studies will both provide valuable insights for further research in this area and guide the use of appropriate methods for clinical cardiac reconstruction from point clouds extracted from MR or CT scans.

A. Data

We conduct the empirical studies using a subset of our collected benchmark, UKBio, while publicly releasing the whole benchmark to facilitate the community research. More specifically, TODO.

B. Evaluation Metrics

For cardiac representation, evaluate the capacity of the model to represent known shapes, i.e. shapes that were in the training set, from only a restricted-size latent code - testing the limit of expressive capability of the representations. — capacity of the latent code, key features with low dimension, capacity of different representation model?

For cardiac reconstruction, evaluate the capacity of the model to represent known shapes. We use chamfer distance (CD) and earth mover distance (EMD) to evaluate reconstruction accuracy. Reconstructed meshes are extracted at a resolution of 256^3 for all methods. CD and EMD are evaluated using 10K and 8K sampled points respectively. Self-intersection?

Dense correspondence uncertainty measurement: show visualization examples to better check the correspondence quality.

1. class-level correspondence visualization
2. point-level measured by the following equation

$$\mu(p_i, p_j) = 1 - \exp(-\gamma \| (p_i + v_i) - (p_j + v_j) \|_2^2) \quad (3)$$

two instances $\mathcal{O}_i$ and $\mathcal{O}_j$, $p_i \in \mathcal{O}_i$, $p_j \in \mathcal{O}_j$; v_i and v_j are corresponding deformations; γ is a scaling factor.

C. Methods and Implementation Details

In section IV, we have categorized existing methods according to It is less feasible to study and empirically compare all the existing methods; instead, we take the strategy of selecting representative ones from each method group, assuming that our studies and conclusions would be generalized in the same groups of existing methods. More specifically, we adopt the most representative learning-free reconstruction methods, DG [2], BPA [3], PSR [4], and RIMLS [5]. These learning-free methods work on single-instance geometric reconstruction, while the learning-based shape representation methods can model a shape distribution. For learning based method, we adopt AtlasNet [6] that reconstructs 3D surfaces by learning a set of parameterized surface elements, DeepSDF [1] which is the first method to learn a continuous SDF for shape representation, IGR [7], DIF-Net [8] and DIT [9] are deformation field-based methods, MR-Net [10], MeshDiffusion [11], NDM [12], 3DShape2VecSet [13], NDF [14] and HNDF [15] represent deformation fields using neural differential flows, but using additional shape-coded DeepSDF to model SDF, CoFie [16], TetraDiffusion [17], SHDF [18]. **to perform a fine-grained classification**

DG [2]:

BPA [3]:

PSR [4]:

RIMLS [5]:

DeepSDF [1]: Shape-coded DeepSDF takes as input both a 3D query point x and a latent code z_i , and outputs an approximate SDF representing the corresponding shape. The latent vector z_i encodes the target shape and is optimized within an auto-decoder framework, enabling the model to represent a wide variety of shapes.

IGR [7]:

DIF-Net [8]: Deformed implicit field (DIF) can model 3D shape of a category and generate dense correspondences among shapes, where a shape latent space, a template implicit field, and a 3D deformation field are jointly learned. produce 3D shapes, and build dense correspondence across different shapes, reflecting shape structure discrepancy

DIT [9]:

MR-Net [10]:

MeshDiffusion [11]:

NDM [12]:

3DShape2VecSet [13]:

NDF [14] and HNDF [15]:

CoFie [16]:

TetraDiffusion [17]:

SHDF [18]:

We implement DG [2] using the the Computational Geometry Algorithms Library (CGAL), and directly execute BPA [3], PSR [4], and RIMLS [5] with proper parameter tunings in MeshLab [19]. For learning-based methods, we use the implementation codes publicly released by the authors, again with proper tuning of hyper-parameters for individual surfaces to be reconstructed. For those without code releasing, we re-implement their algorithms and tune the respective hyper-parameters as the optimal ones. **Networks are all implemented as MLPs, or try other architectures? All parameters are trained end-to-end using Adam optimizer.**

D. Experiments

1) **Data Efficiency Study:** We evaluate different baseline methods by varying both the sampling strategy and the number of points per training sample. Sampling strategies include **uniform, random, and farthest point sampling (FPS)**, while the number of sampled points is set to **500000, 300000, or 100000**. This design allows a systematic comparison of the performance of each method under different training configurations.

2) **Shape Code Representation Study:** We evaluate different shape code representations by varying both the latent code type and the encoding capacity. Latent code configurations include global latent codes **with dimensions of 16, 64, 128, 256**, local latent codes **with spatial resolutions of 4×4×4, 8×8×8, 16×16×16 grid partitions (per-cell dimension fixed to 16)**, and structured latent codes (e.g., triplane representation) with feature map resolutions of 32×32, 64×64, 128×128 and channel dimensions of 16, 32, 64. This design allows a systematic comparison of reconstruction fidelity and shape interpolation quality under different shape encoding schemes.

3) **Template learning??? (point correspondence) for registration:**

VII. EXPERIMENTAL RESULTS

A. 3D Cardiac Representation

B. 3D Cardiac Reconstruction

C. Shape Registration

VIII. CONCLUSION

The conclusion goes here.

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This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

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IX. BIOGRAPHY SECTION

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